AP Calculus BC
Unit 2 Review Book

Differentiation: Definition and Fundamental Properties

This review book is designed for a **full beginner-friendly final review**. It covers every topic in AP Calculus BC Unit 2 in a way that is detailed, organized, and fast to study from. Every major formula is shown in rendered mathematics, each lesson highlights common traps, and each section includes representative AP-style examples.

Course Unit:	Unit 2
Scope:	Definition of the derivative, differentiability, core derivative rules, and foundational derivative formulas
Best Use:	Read section by section, then return to the summary sheet at the end for memorization and speed review

Prepared for AP Calculus BC review

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How to Use This Book

This book is built for **mastery and speed**. For each lesson, focus on four questions:

1. What does the concept actually mean?
2. What formula or rule must I know cold?
3. What does AP usually ask about this topic?
4. What mistakes do students make under time pressure?

A strong Unit 2 student can do all of the following without hesitation:

- interpret a derivative as both a **tangent line slope** and an **instantaneous rate of change**,
- compute a derivative from the **limit definition**,
- estimate a derivative from a **graph or table**,
- determine where a derivative **does not exist**,
- apply the **power rule, basic derivative rules, product rule, quotient rule**,
- and differentiate the core transcendental and trigonometric functions quickly and accurately.

1. Unit 2 Big Picture

Unit 2 turns the idea of a limit into one of the most important objects in calculus: the **derivative**. In plain language, the derivative measures how fast a quantity is changing at an exact moment. In geometry, it is the slope of the tangent line. In context, it is an instantaneous rate of change.

The two anchor ideas of the entire unit are:

$$\text{derivative} = \text{instantaneous rate of change}$$

and

$$\text{derivative} = \text{slope of the tangent line.}$$

Once these ideas are secure, the rest of Unit 2 is about learning how to compute derivatives efficiently and how to interpret them correctly.

2. 2.1 Defining Average and Instantaneous Rates of Change at a Point

2.1 Core idea

Average rate of change tells you how much a function changes *on an interval*. Instantaneous rate of change tells you how much a function is changing *at one exact point*.

2.2 Essential formulas

For a function f on the interval $[a, b]$, the average rate of change is

$$\frac{f(b) - f(a)}{b - a}.$$

This is the slope of the secant line through $(a, f(a))$ and $(b, f(b))$.

At $x = a$, the instantaneous rate of change is defined by the limit

$$\lim_{h \to 0} \frac{f(a+h) - f(a)}{h}.$$

This is the slope of the tangent line at $x = a$.

2.3 What AP asks

- Distinguish average rate of change from instantaneous rate of change.
- Estimate the instantaneous rate of change numerically from smaller and smaller intervals.
- Interpret a derivative in context with correct units.

Common trap: A secant slope is *not* automatically a tangent slope. The secant slope only approximates the derivative when the two x-values become extremely close.

Example 2.1A. Suppose

$$\frac{f(2.1) - f(2)}{0.1} = 5.7, \qquad \frac{f(2.01) - f(2)}{0.01} = 5.97, \qquad \frac{f(2.001) - f(2)}{0.001} = 5.997.$$

Estimate $f'(2)$.

Solution. These are difference quotients of the form

$$\frac{f(2+h) - f(2)}{h}.$$

As $h \to 0$, the values approach 6. Therefore,

$$f'(2) \approx 6.$$

Example 2.1B. A car's position is given by $s(t)$ in meters, where t is in seconds. Explain the meaning of $s'(4) = 12$.

Solution. At $t = 4$ seconds, the car's position is changing at an instantaneous rate of 12 meters per second. In this context, $s'(4)$ is the car's instantaneous velocity at $t = 4$.

3. 2.2 Defining the Derivative of a Function and Using Derivative Notation

3.1 Core idea

The derivative is the formal calculus object that records instantaneous change. The derivative can be defined in two equivalent standard ways.

3.2 Two limit definitions

At a point $x = a$,

$$f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}$$

and equivalently

$$f'(a) = \lim_{x \to a} \frac{f(x) - f(a)}{x - a}.$$

As a function,

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}.$$

3.3 Derivative notation

All of the following may refer to the derivative:

$$f'(x), \qquad y', \qquad \frac{dy}{dx}, \qquad \frac{d}{dx}[f(x)].$$

At a specific point, you may see

$$f'(a) \qquad \text{or} \qquad \left.\frac{dy}{dx}\right|_{x=a}.$$

3.4 Meaning review

- Algebraically: a limit of difference quotients.
- Geometrically: slope of the tangent line.
- In context: instantaneous rate of change.

Common trap: Students often confuse $f'(x)$ with $f(x)$. The derivative is a *new function* that tells how the original function changes.

Example 2.2A. What derivative is represented by

$$\lim_{h \to 0} \frac{(3+h)^4 - 81}{h}?$$

Solution. Let $f(x) = x^4$. Then

$$f(3+h) = (3+h)^4, \qquad f(3) = 3^4 = 81.$$

So the limit is

$$\lim_{h \to 0} \frac{f(3+h) - f(3)}{h} = f'(3).$$

Example 2.2B. Use the definition of the derivative to compute $f'(x)$ for $f(x) = x^2$.
Solution.

$$f'(x) = \lim_{h\to 0} \frac{(x+h)^2 - x^2}{h}.$$

Expand the numerator:

$$(x+h)^2 - x^2 = x^2 + 2xh + h^2 - x^2 = 2xh + h^2.$$

So

$$f'(x) = \lim_{h\to 0} \frac{2xh + h^2}{h} = \lim_{h\to 0}(2x + h) = 2x.$$

Therefore,

$$f'(x) = 2x.$$

4. 2.3 Estimating Derivatives of a Function at a Point

4.1 Core idea

Sometimes AP will not give you a formula for $f(x)$. Instead, you may have a graph or a table. You still need to estimate the derivative by thinking about tangent line slope or nearby difference quotients.

4.2 From a graph

To estimate $f'(a)$ from a graph:

1. Locate the point $x = a$.
2. Picture or sketch the tangent line.
3. Estimate its slope using two readable points on that tangent line.

4.3 From a table

Use values very close to a and compute

$$\frac{f(a+h) - f(a)}{h} \quad \text{or} \quad \frac{f(a) - f(a-h)}{h}.$$

If both sides agree, you have a strong estimate.

Common trap: The size of the function value has nothing to do with the size of the derivative. A point can have a large y-value and still have slope 0, or a small y-value and still have a steep slope.

Example 2.3A. The table below gives values of f near $x = 1$.

x	0.9	0.99	1.00	1.01	1.1
$f(x)$	2.19	2.0199	2	1.9801	1.79

Estimate $f'(1)$.
Solution. Using the right-hand difference quotient,

$$\frac{f(1.01) - f(1)}{1.01 - 1} = \frac{1.9801 - 2}{0.01} = -1.99.$$

Using the left-hand difference quotient,

$$\frac{f(1) - f(0.99)}{1 - 0.99} = \frac{2 - 2.0199}{0.01} = -1.99.$$

So a good estimate is

$$f'(1) \approx -2.$$

5. 2.4 Connecting Differentiability and Continuity - Determining When Derivatives Do and Do Not Exist

5.1 Core idea

If a function is differentiable at a point, then it is automatically continuous there. But the reverse is not always true.

$$\text{Differentiable at } x = a \implies \text{Continuous at } x = a.$$

But

$$\text{Continuous at } x = a \not\Longrightarrow \text{Differentiable at } x = a.$$

5.2 When derivatives fail to exist

A derivative may not exist at a point because of:

- a discontinuity,
- a sharp corner,
- a cusp,
- a vertical tangent,
- wild oscillation.

5.3 What to check on AP

When AP asks whether $f'(a)$ exists, think:

1. Is the function continuous at a?
2. Is there a corner or cusp?
3. Is the tangent vertical?

4. Do the one-sided slopes agree?

Example 2.4A. Does $f(x) = |x|$ have a derivative at $x = 0$?
Solution. The graph is continuous at $x = 0$, but the left-hand slope is -1 and the right-hand slope is 1. Since the one-sided slopes are not equal, $f'(0)$ does not exist.

Example 2.4B. Does $f(x) = x^{1/3}$ have a derivative at $x = 0$?
Solution. Its derivative for $x \neq 0$ is

$$f'(x) = \frac{1}{3x^{2/3}}.$$

As $x \to 0$, the slope becomes unbounded, so the tangent is vertical. Therefore, $f'(0)$ does not exist.

Common trap: Students often think "continuous" means "differentiable." That is false. The classic counterexample is $|x|$ at $x = 0$.

6. 2.5 Applying the Power Rule

6.1 Core idea

The power rule is the first major computational shortcut for derivatives.

6.2 Power rule

For any real number n for which the derivative makes sense,

$$\frac{d}{dx}(x^n) = nx^{n-1}.$$

This includes positive integer powers, negative powers, and many fractional powers.

6.3 Examples of the rule

$$\frac{d}{dx}(x^5) = 5x^4$$

$$\frac{d}{dx}(x^{-3}) = -3x^{-4}$$

$$\frac{d}{dx}(x^{1/2}) = \frac{1}{2}x^{-1/2} = \frac{1}{2\sqrt{x}}$$

6.4 AP focus

- Differentiate polynomials quickly.
- Rewrite roots and rational expressions using exponents before differentiating.
- Avoid sign errors with negative exponents.

Example 2.5A. Differentiate

$$f(x) = 4x^6 - 3x^{-2} + 7\sqrt{x}.$$

Solution. Rewrite $\sqrt{x}$ as $x^{1/2}$:

$$f(x) = 4x^6 - 3x^{-2} + 7x^{1/2}.$$

Then

$$f'(x) = 24x^5 + 6x^{-3} + \frac{7}{2}x^{-1/2}.$$

You may also write the last term as

$$\frac{7}{2\sqrt{x}}.$$

Common trap: For x^{-2}, the derivative is

$$-2x^{-3},$$

not

$$-2x^{-1}.$$

Both the coefficient and the exponent must be updated correctly.

7. 2.6 Derivative Rules - Constant, Sum, Difference, and Constant Multiple

7.1 Core idea

These rules let you break a complicated derivative into manageable parts.

7.2 Rules to memorize

$$\frac{d}{dx}(c) = 0$$

$$\frac{d}{dx}[f(x) + g(x)] = f'(x) + g'(x)$$

$$\frac{d}{dx}[f(x) - g(x)] = f'(x) - g'(x)$$

$$\frac{d}{dx}[cf(x)] = cf'(x)$$

7.3 Why these matter

They let you combine the power rule with linear combinations of functions very quickly.

Example 2.6A. Differentiate

$$y = 9 - 5x^4 + 2x^{-3} + 8x.$$

Solution.

$$y' = 0 - 20x^3 - 6x^{-4} + 8.$$

So

$$y' = -20x^3 - 6x^{-4} + 8.$$

Common trap: Constant terms always differentiate to 0. Under time pressure, students often leave them in place by mistake.

8. 2.7 Derivatives of $\cos x$, $\sin x$, e^x, and $\ln x$

8.1 Core formulas to memorize exactly

$$\frac{d}{dx}(\sin x) = \cos x$$
$$\frac{d}{dx}(\cos x) = -\sin x$$
$$\frac{d}{dx}(e^x) = e^x$$
$$\frac{d}{dx}(\ln x) = \frac{1}{x} \quad (x > 0)$$

8.2 What makes these special

These are not derived by the power rule. They are new fundamental derivative formulas and appear constantly throughout AP Calculus.

8.3 AP expectations

You should be able to use these formulas immediately, combine them with earlier rules, and evaluate derivatives at specific points.

Example 2.7A. Differentiate

$$f(x) = \sin x + \cos x + e^x + \ln x.$$

Solution.

$$f'(x) = \cos x - \sin x + e^x + \frac{1}{x}.$$

Example 2.7B. Find the slope of the tangent line to

$$y = \sin x + e^x$$

at $x = 0$.
Solution.

$$y' = \cos x + e^x.$$

Then

$$y'(0) = \cos 0 + e^0 = 1 + 1 = 2.$$

So the tangent slope is

$$2.$$

High-frequency error list:

- $(\cos x)'$ is **not** $\sin x$; it is $-\sin x$.
- $(e^x)' = e^x$, not xe^{x-1}.
- $(\ln x)' = 1/x$, not $1/\ln x$.

9. 2.8 The Product Rule

9.1 Core idea

The derivative of a product is *not* the product of the derivatives. This is one of the most common conceptual mistakes in first-semester calculus.

9.2 Product rule formula

If

$$y = f(x)g(x),$$

then

$$y' = f'(x)g(x) + f(x)g'(x).$$

A memory phrase is:

$$\text{first derivative times second} + \text{first times second derivative.}$$

9.3 AP focus

The product rule is commonly mixed with:

- power rule,
- trigonometric derivatives,
- exponential derivatives,
- logarithmic derivatives.

Example 2.8A. Differentiate

$$y = x^2 \sin x.$$

Solution. Let $f(x) = x^2$ and $g(x) = \sin x$. Then

$$f'(x) = 2x, \qquad g'(x) = \cos x.$$

So

$$y' = 2x \sin x + x^2 \cos x.$$

Example 2.8B. Differentiate

$$y = (x^3 + 1)e^x.$$

Solution.

$$y' = (3x^2)e^x + (x^3 + 1)e^x.$$

You may factor out e^x:

$$y' = e^x(3x^2 + x^3 + 1).$$

Common trap:

$$(fg)' \neq f'g'.$$

That incorrect rule appears often under pressure, especially on multiple choice.

10. 2.9 The Quotient Rule

10.1 Core idea

The derivative of a quotient also requires its own formula. It is not valid to differentiate numerator and denominator separately and divide.

10.2 Quotient rule formula

If

$$y = \frac{f(x)}{g(x)},$$

then

$$y' = \frac{f'(x)g(x) - f(x)g'(x)}{[g(x)]^2}.$$

A memory phrase is:

low d-high minus high d-low over low squared.

Example 2.9A. Differentiate

$$y = \frac{x^2+1}{x-1}.$$

Solution. Take

$$f(x) = x^2+1, \qquad g(x) = x-1.$$

Then

$$f'(x) = 2x, \qquad g'(x) = 1.$$

So

$$y' = \frac{2x(x-1) - (x^2+1)(1)}{(x-1)^2}.$$

Simplifying the numerator gives

$$y' = \frac{2x^2 - 2x - x^2 - 1}{(x-1)^2} = \frac{x^2 - 2x - 1}{(x-1)^2}.$$

Example 2.9B. Differentiate

$$y = \frac{\sin x}{x}.$$

Solution.

$$y' = \frac{x\cos x - \sin x}{x^2}.$$

Common traps:

- Reversing the order in the numerator.
- Forgetting to square the denominator.
- Writing
$$\left(\frac{f}{g}\right)' = \frac{f'}{g'}$$
which is false.

11. 2.10 Finding the Derivatives of Tangent, Cotangent, Secant, and/or Cosecant Functions

11.1 Core formulas to memorize

$$\frac{d}{dx}(\tan x) = \sec^2 x$$
$$\frac{d}{dx}(\cot x) = -\csc^2 x$$
$$\frac{d}{dx}(\sec x) = \sec x \tan x$$
$$\frac{d}{dx}(\csc x) = -\csc x \cot x$$

11.2 Why these matter

These derivatives appear often in multiple-choice questions because they are easy to confuse. They also pair naturally with the product rule and quotient rule.

Example 2.10A. Differentiate
$$y = \sec x + \csc x + \tan x.$$
Solution.
$$y' = \sec x \tan x - \csc x \cot x + \sec^2 x.$$

Example 2.10B. Differentiate
$$y = x \tan x.$$
Solution. Use the product rule:
$$y' = 1 \cdot \tan x + x \cdot \sec^2 x.$$
So
$$y' = \tan x + x \sec^2 x.$$

High-frequency error list:

- $(\tan x)'$ is $\sec^2 x$, not $\sec x$.
- $(\sec x)'$ is $\sec x \tan x$, not $\tan^2 x$.
- Both $(\cot x)'$ and $(\csc x)'$ include a negative sign.

12. Mixed AP-Style Practice

Problem 1

Use the derivative definition to compute the derivative of

$$f(x) = 3x^2 - 2x.$$

Solution.

$$f'(x) = \lim_{h\to 0} \frac{3(x+h)^2 - 2(x+h) - (3x^2 - 2x)}{h}.$$

Expand:

$$3(x+h)^2 - 2(x+h) = 3x^2 + 6xh + 3h^2 - 2x - 2h.$$

Subtract $3x^2 - 2x$:

$$6xh + 3h^2 - 2h.$$

Divide by h:

$$f'(x) = \lim_{h\to 0}(6x + 3h - 2) = 6x - 2.$$

Problem 2

Find all values of x where the derivative of

$$f(x) = |x - 3|$$

does not exist.

Solution. The graph has a corner where the inside equals 0, namely at

$$x = 3.$$

So the derivative does not exist at

$$x = 3.$$

Problem 3

Differentiate

$$y = (x^2 + 1)(\cos x) + \frac{e^x}{x}.$$

Solution. For the first term, use the product rule:

$$\frac{d}{dx}[(x^2 + 1)\cos x] = 2x\cos x - (x^2 + 1)\sin x.$$

For the second term, use the quotient rule:

$$\frac{d}{dx}\left(\frac{e^x}{x}\right) = \frac{xe^x - e^x}{x^2} = \frac{e^x(x-1)}{x^2}.$$

Therefore,

$$y' = 2x\cos x - (x^2 + 1)\sin x + \frac{e^x(x-1)}{x^2}.$$

Problem 4

A particle moves along a line with position $s(t)$ in feet. Explain the meaning of

$$s'(5) = -7.$$

Solution. At time $t = 5$, the particle's position is changing at an instantaneous rate of -7 feet per second. The negative sign indicates motion in the negative direction.

13. Unit 2 Formula Sheet

Definition of derivative

$$f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}$$

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

Basic algebraic rules

$$\frac{d}{dx}(c) = 0$$

$$\frac{d}{dx}(x^n) = nx^{n-1}$$

$$\frac{d}{dx}[cf(x)] = cf'(x)$$

$$\frac{d}{dx}[f(x) \pm g(x)] = f'(x) \pm g'(x)$$

Core transcendental and trigonometric derivatives

$$(\sin x)' = \cos x$$

$$(\cos x)' = -\sin x$$

$$(e^x)' = e^x$$

$$(\ln x)' = \frac{1}{x}$$

$$(\tan x)' = \sec^2 x$$

$$(\cot x)' = -\csc^2 x$$

$$(\sec x)' = \sec x \tan x$$

$$(\csc x)' = -\csc x \cot x$$

Product rule

$$(fg)' = f'g + fg'$$

Quotient rule

$$\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$$

14. Final Unit 2 Checklist

Before you move on, make sure you can do each of these quickly and accurately.

Skill	Ready?
Use the limit definition of the derivative for simple polynomial functions	☐
Interpret a derivative as tangent slope and instantaneous rate of change	☐
Estimate a derivative from a graph or a table	☐
Determine where a derivative does not exist and explain why	☐
Apply the power rule accurately with positive, negative, and fractional exponents	☐
Differentiate sums, differences, constant multiples, and constants rapidly	☐
Recall and apply derivatives of $\sin x$, $\cos x$, e^x, and $\ln x$ without hesitation	☐
Recall and apply derivatives of $\tan x$, $\cot x$, $\sec x$, and $\csc x$ without hesitation	☐
Use the product rule correctly	☐
Use the quotient rule correctly	☐
Distinguish average rate of change from instantaneous rate of change	☐

Best final advice for Unit 2. If you feel shaky, return to four things first: the derivative definition, differentiability vs. continuity, the product rule, and the quotient rule. Those four ideas generate a large percentage of early-course AP Calculus errors.